Planar diagram algebra and 3D TQFT

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1 Planar diagram algebra

1.1 Unitary fusion category

1.1.1 Definition

Here we discuss the definition of *unitary fusion category*, which is central in three dimensional topological quantum field theories (3D TQFTs). Each step adds additional algebraic structure.

Monoidal Category.

Definition 1.1. *A monoidal category is a tuple* $(\mathsf{C}, \otimes, 1, \alpha, \lambda, \rho)$ *where:*

- $\otimes : \mathsf{C} \times \mathsf{C} \to \mathsf{C}$ *is a bifunctor,*
- α *(associator),* λ *(left unit), and* ρ *(right unit) are natural isomorphisms*

$$\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z),$$

$$\lambda_X : 1 \otimes X \to X, \qquad \rho_X : X \otimes 1 \to X,$$

satisfying the pentagon and triangle coherence axioms.

Enriched Category. Let $(\mathsf{E}, \otimes, 1)$ be a monoidal category.

Definition 1.2. *A* E*-enriched category* C *consists of:*

- *A class of objects* $\mathrm{Ob}(\mathsf{C})$*,*
- *For each pair* X, Y*, an object*
$$\mathrm{Hom}_{\mathsf{C}}(X, Y) \in \mathsf{E},$$
- *Composition morphisms in* E*:*
$$\mathrm{Hom}_{\mathsf{C}}(Y, Z) \otimes \mathrm{Hom}_{\mathsf{C}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathsf{C}}(X, Z),$$
- *Unit morphisms*
$$1_{\mathsf{E}} \to \mathrm{Hom}_{\mathsf{C}}(X, X),$$

such that associativity and unit axioms hold inside E*.*

Example 1.3. *Ordinary categories correspond to enrichment over* Set*.*

Linear Category.

Definition 1.4. *A* k*-linear category (or linear category over a field* k*)* C *is a category enriched over* Vect_k*.*

Equivalently:

- Each $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a k-vector space,
- Composition is bilinear,
- Unit morphisms are k-linear maps $k \to \mathrm{Hom}_{\mathsf{C}}(X, X)$.

If $\dim_k \mathrm{Hom}_{\mathsf{C}}(X, Y) < \infty$ for all X, Y, the category is *locally finite*.

Remark 1.5. *We introduced monoidal categories* $(\mathsf{E}, \otimes_{\mathsf{E}}, 1_{\mathsf{E}})$ *to define enriched categories. The defining conditions of monoidal category are precisely what we must demand to use the objects as Hom spaces of another category* C*. Below, we will also demand that the* E*-enriched category* $(\mathsf{C}, \otimes_{\mathsf{C}}, 1_{\mathsf{C}})$ *itself be monoidal.*

Rigidity.

Definition 1.6. *A monoidal category* C *is rigid if every object* X *admits a (left and right) dual* X^* *with morphisms*
$$\mathrm{ev}_X : X^* \otimes X \to 1, \qquad \mathrm{coev}_X : 1 \to X \otimes X^*$$
such that the zig–zag identities hold:
$$(\mathrm{id}_X \otimes \mathrm{ev}_X) \circ (\mathrm{coev}_X \otimes \mathrm{id}_X) = \mathrm{id}_X, \tag{1a}$$
$$(\mathrm{ev}_X \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X) = \mathrm{id}_{X^*}. \tag{1b}$$
and similarly for X^**.*

Rigidity ensures that categorical traces and dimensions can be defined.

Semisimplicity.

Definition 1.7. *A k-linear category C is semisimple if:*

- *Every object is a finite direct sum of simple objects,*
- $\mathrm{End}_{\mathsf{C}}(X) \simeq k$ *for every simple object X,*
- $\mathrm{Hom}_{\mathsf{C}}(X, Y) = 0$ *for non-isomorphic simple objects X, Y.*

In a semisimple k-linear category, every object uniquely decomposes as a finite sum

$$X \simeq \bigoplus_i X_i^{\oplus n_i}$$

with X_i simple.

Fusion Category.

Definition 1.8. *A fusion category over an algebraically closed field k is a k-linear monoidal category C such that:*

- C *is semisimple,*
- *It has finitely many isomorphism classes of simple objects,*
- $\dim_k \mathrm{Hom}_{\mathsf{C}}(X, Y) < \infty$ *for all X, Y,*
- *The unit object 1 is simple,*
- C *is rigid.*

Thus a fusion category is a finite, semisimple, rigid, linear monoidal category. Fusion rules are given by

$$X_i \otimes X_j \simeq \bigoplus_k N_{ij}^k X_k, \qquad N_{ij}^k \in \mathbb{Z}_{\geq 0}. \tag{2}$$

Unitary Fusion Category. Let $k = \mathbb{C}$.

Definition 1.9. *A unitary fusion category is a fusion category C over $\mathbb{C}$ equipped with:*

- *A $*$-structure making each $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ a Hilbert space,*
- *Composition compatible with the $*$-operation,*
- *The tensor product compatible with $*$,*
- *The associator α and units λ, ρ are unitary.*

Equivalently, C is a $\mathbb{C}$-linear fusion category enriched over finite-dimensional Hilbert spaces. In this case:

- Categorical dimensions are positive real numbers,
- Duality maps can be chosen unitary,
- The theory admits a consistent Hermitian diagrammatic calculus.

1.2 Fusion category and planar diagrams

String Diagram Calculus. Let C be a monoidal category. Morphisms in C admit a graphical representation by *string diagrams*:

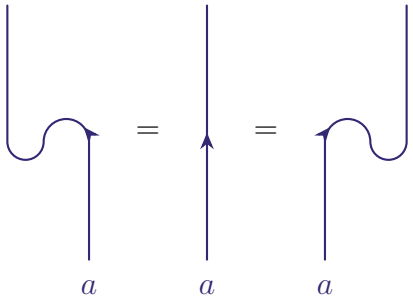
- Objects are labeled strands.
- Morphisms $f : X \to Y$ are boxes with input strand X and output strand Y.
- Vertical stacking represents composition.
- Horizontal juxtaposition represents tensor product.

Mac Lane's coherence theorem implies that all diagrams built using associators and units are invariant under planar isotopy. Thus, a monoidal category is equivalently described by planar diagrammatic calculus.

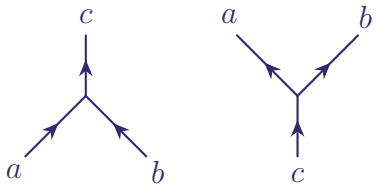
Example 1.10. *If C is rigid (i.e. each object has dual), the evaluation/coevaluation maps are represented diagrammatically by cups and caps.*



The zig–zag identities, Eq. (1), become planar isotopy relations between a straight line and a bent line. Diagrammatically, this allows strands to bend. Hence a rigid monoidal category corresponds to planar diagrams with oriented strands and allowed bending.

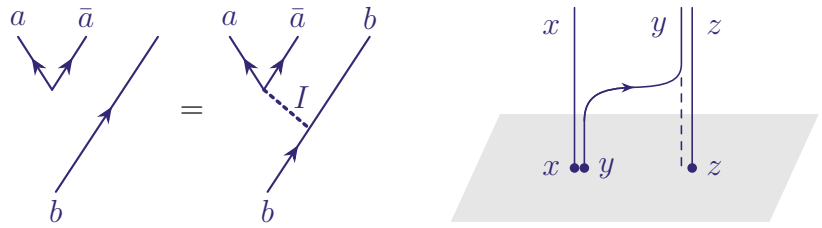


Example 1.11. *If C is a fusion category, every fusion can be represented by a sum of trivalent vertices.*



A single vertex suffices if the fusion rule is multiplicity-free, i.e. N_{ij}^k takes only two values $0, 1$ in Eq. (2). Otherwise, a fusion must be interpreted as a sum ("quantum superposition") of

vertices. The tensor unit can be represented by an "invisible line", and an associator is an H-shape diagram.



A fusion category is k-linear, semisimple, rigid, and has finitely many simple objects. Semisimplicity implies that morphisms between tensor products decompose into finite sums over simple labels. Therefore, any diagram whose boundary is labeled by general objects can be rewritten as a finite direct sum of diagrams whose strands are labeled only by simple objects. An example of such diagram is depicted in Fig. 1. For simple objects $\{X_i\}$, fusion rules are encoded diagrammatically by trivalent vertices:

$$X_i \otimes X_j \simeq \bigoplus_k N_{ij}^k X_k.$$

Associativity is implemented by F-moves, which correspond to local planar re-coupling transformations. The pentagon identity ensures consistency of planar re-association moves.

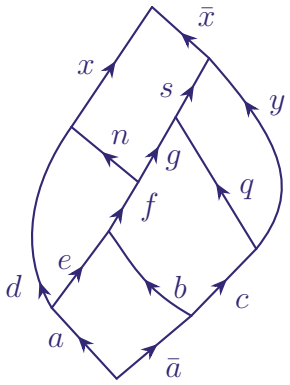


Figure 1: A planar fusion diagram starting and ending with the vacuum (tensor unit).

Endomorphism Algebras as Diagram Algebras. Fix an object X. The spaces

$$\mathrm{Hom}(X^{\otimes n}, X^{\otimes m})$$

are finite-dimensional, and $\mathrm{End}(X^{\otimes n})$ are finite-dimensional algebras. Elements are represented by planar diagrams with n input and m output boundary points. Composition is vertical stacking; tensor product is horizontal concatenation. Thus a fusion category produces a family of planar diagram algebras.

Conceptual Summary.

- A monoidal category = planar tensor calculus.

- Rigidity = planar diagrams with cups and caps.
- Semisimplicity = finite label set and finite diagram evaluation.
- Fusion category = finite, rigid planar diagram theory.

Therefore, a fusion category can be regarded as the algebraic formalization of a consistent planar diagrammatic theory with finitely many particle labels and local recoupling rules.

1.3 Braiding and twist

1.3.1 Braided Fusion Category

We assume throughout that C is a fusion category over an algebraically closed field k (typically $k = \mathbb{C}$). Certain fusion categories have additional structure beyond the fusion structure.

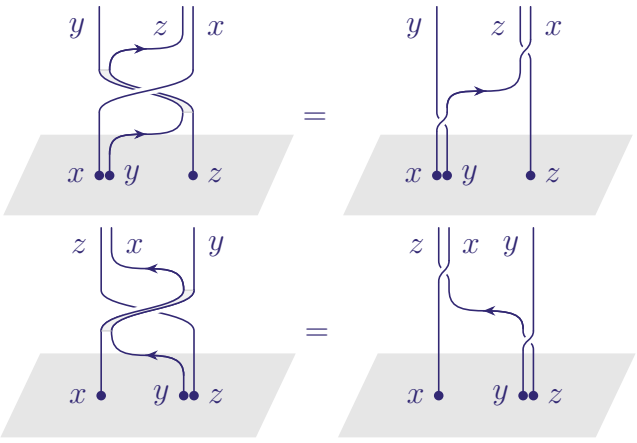
Definition 1.12. *A braiding on a monoidal category C is a natural family of isomorphisms*

$$c_{X,Y} : X \otimes Y \longrightarrow Y \otimes X \tag{3}$$

satisfying the two hexagon axioms:

$$\alpha_{Y,Z,X} \circ c_{X,Y\otimes Z} \circ \alpha_{X,Y,Z} = (\mathrm{id}_Y \otimes c_{X,Z}) \circ \alpha_{Y,X,Z} \circ (c_{X,Y} \otimes \mathrm{id}_Z), \tag{4a}$$

$$\alpha^{-1}_{Z,X,Y} \circ c_{X\otimes Y,Z} \circ \alpha^{-1}_{X,Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y) \circ \alpha^{-1}_{X,Z,Y} \circ (\mathrm{id}_X \otimes c_{Y,Z}). \tag{4b}$$



Definition 1.13. *A braided fusion category is a fusion category equipped with a braiding $c_{X,Y}$.*

The braiding provides a coherent way to "swap" tensor factors. Diagrammatically, $c_{X,Y}$ corresponds to a crossing of strands. The hexagon axioms ensure compatibility between braiding and associativity. If additionally

$$c_{Y,X} \circ c_{X,Y} = \mathrm{id}_{X\otimes Y} \tag{5}$$

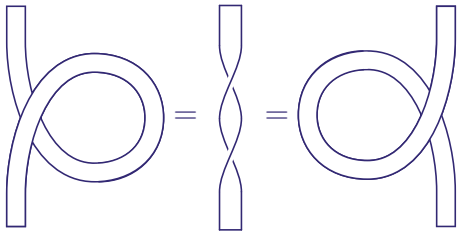
for all X, Y, the braiding is *symmetric*, and C is a symmetric fusion category.

1.3.2 Ribbon category

To define a ribbon category, we first introduce a twist.

Definition 1.14. *Let* $(\mathsf{C}, \otimes, 1)$ *be a braided rigid monoidal category. A* <u>twist</u> *is a natural isomorphism* $\theta_X : X \longrightarrow X$ *such that*

- $\theta_1 = \mathrm{id}_1$,
- $\theta_{X\otimes Y} = c_{Y,X} \circ c_{X,Y} \circ (\theta_X \otimes \theta_Y)$.



The twist measures the failure of double braiding to be trivial. Diagrammatically, θ_X corresponds to a full 2π twist of a strand (ribbon).

Definition 1.15. *A* <u>ribbon category</u> *(or balanced rigid braided category) is a rigid braided monoidal category equipped with a twist* θ *such that the twist is compatible with duality:*

$$\theta_{X^*} = (\theta_X)^*.$$

Equivalently, the left and right traces defined using the braiding and twist coincide; hence ribbon categories are automatically spherical.

Definition 1.16. *A* <u>ribbon fusion category</u> *is a fusion category equipped with a braiding and a compatible twist making it a ribbon category.*

Structure Summary.

$$\text{Fusion} \;\Rightarrow\; \text{Braided Fusion} \;\Rightarrow\; \text{Ribbon Fusion}. \tag{6}$$

- Fusion category: planar tensor calculus.
- Braided fusion category: crossings allowed.
- Ribbon fusion category: crossings plus consistent 2π twists, enabling framed link invariants.

Ribbon fusion categories provide the algebraic input for 3-dimensional topological quantum field theories and link invariants. If the braiding is additionally non-degenerate, one obtains a *modular tensor category*.

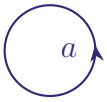
2 State-sum TQFTs

2.1 Quantum dimension and F-matrix

The quantum dimension d_X and F-matrix F^{bad}_{ecf} could be defined when we introduced fusion categories, but we do it here as they are the input data of a Turaev–Viro state-sum TQFT.

2.1.1 Quantum dimension

Let C be a fusion category. In the language of planar diagram algebra, the *quantum dimension* of an object $X \in \mathsf{C}$ is defined by a loop labeled by X.



$$= d_a = d_{\bar{a}}$$

It is always true that $d_I = 1$, meaning that loops of vacuum can be freely added or removed from a diagram. Another general relation is

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad N_{ab}^c := \dim \mathrm{Hom}(a \otimes b, c). \tag{7}$$

Using this, we can prove the following.

Proposition 2.1. $\sum_{a,b} N_{ab}^c d_a d_b = \mathcal{D}^2 d_c$.

Proof. By rigidity, we have $N_{ab}^c = N_{\bar{c}a}^{\bar{b}}$. Hence

$$\sum_b N_{ab}^c d_b = \sum_b N_{\bar{c}a}^{\bar{b}} d_b = \sum_d N_{\bar{c}a}^d d_d = d_{\bar{c}} d_a = d_c d_a$$

$$\Rightarrow \sum_{a,b} N_{ab}^c d_a d_b = \sum_a \left(d_a \sum_b N_{ab}^c d_b \right) = \sum_a d_a^2 d_c = \mathcal{D}^2 d_c.$$

□

2.1.2 F-matrix

Let C be a fusion category over an algebraically closed field k with simple objects $\{X_i\}_{i \in I}$. For simple objects X_i, X_j, X_k, define the fusion multiplicities

$$N_{ij}^k := \dim_k \mathrm{Hom}(X_i \otimes X_j, X_k).$$

Fix, once and for all, a basis

$$\{ t_{ij}^{k,\alpha} : X_i \otimes X_j \to X_k \}_{\alpha=1}^{N_{ij}^k}.$$

By duality, this also gives a basis of $\mathrm{Hom}(X_k, X_i \otimes X_j)$. The associator in C is the natural isomorphism

$$\alpha_{X_i, X_j, X_k} : (X_i \otimes X_j) \otimes X_k \longrightarrow X_i \otimes (X_j \otimes X_k).$$

Because the category is semisimple, both sides decompose into direct sums of simple objects. In particular, $\mathrm{Hom}((X_i \otimes X_j) \otimes X_k, X_\ell)$ and $\mathrm{Hom}(X_i \otimes (X_j \otimes X_k), X_\ell)$ are finite-dimensional vector spaces. The F-matrix describes the change of basis between the two ways of decomposing the triple tensor product: $(X_i \otimes X_j) \otimes X_k$ and $X_i \otimes (X_j \otimes X_k)$.

Definition 2.2. *The associator induces linear isomorphisms*

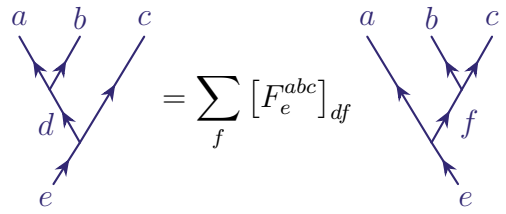
$$F_\ell^{ijk} : \bigoplus_m \mathrm{Hom}(X_i \otimes X_j, X_m) \otimes \mathrm{Hom}(X_m \otimes X_k, X_\ell) \longrightarrow \bigoplus_n \mathrm{Hom}(X_j \otimes X_k, X_n) \otimes \mathrm{Hom}(X_i \otimes X_n, X_\ell).$$

After choosing bases of all fusion spaces, this linear isomorphism is represented by a matrix

$$(F_\ell^{ijk})_{(m\alpha\beta)(n\mu\nu)} \in k.$$

These matrices are called the F-matrices (or $6j$-symbols).

For simplicity, assume that C is multiplicity-free, i.e., $N_{ij}^k = 0, 1$. Then the fusion spaces $\mathrm{Hom}(X_i \otimes X_j, X_k)$ are one dimensional, and there is only one fusion morphism in each fusion channel. Diagrammatically, the F-matrices of a multiplicity-free fusion category are

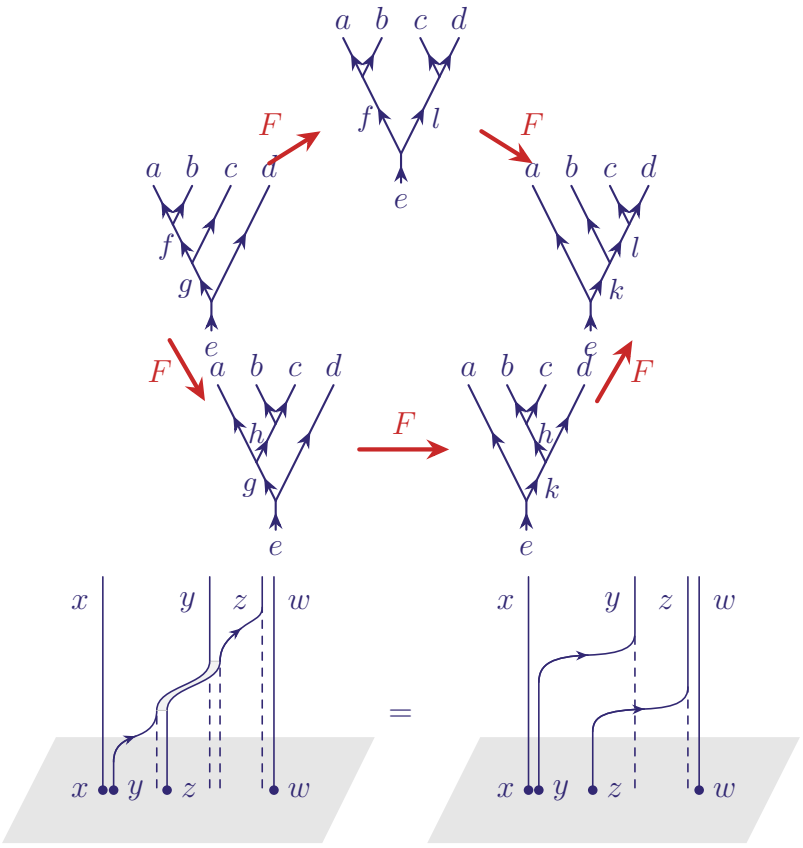


For more general N_{ij}^k, F is simply the matrix form of the associator in the basis determined by chosen fusion morphisms.

The pentagon axiom for the associator translates into the *pentagon equation* for F-matrices. Hence, the F-matrices satisfy a nonlinear polynomial identity ensuring consistency of iterated

re-couplings. For a multiplicity-free fusion category,

$$[F_e^{fcd}]_{gl}[F_e^{abl}]_{fk} = \sum_h [F_g^{abc}]_{fh}[F_e^{ahd}]_{gk}[F_k^{bcd}]_{hl}. \tag{8}$$



Conceptual Summary.

- The associator α is a natural isomorphism.
- Choosing bases of fusion spaces converts α into numerical matrices.
- These matrices are the F-symbols.
- The pentagon axiom becomes the pentagon equation for F.

Thus, F-matrices are nothing more than the coordinate expression of the associator in a fusion category.

2.2 Turaev–Viro

2.2.1 Triangulation

A Turaev–Viro theory is a general procedure which produces a three dimensional (3D) TQFT, which encodes 3-manifold invariants. To assign a partition function $Z(\mathcal{M})$ to a 3-manifold $\mathcal{M}$, a Turaev–Viro type theory *triangulates* $\mathcal{M}$, i.e., decompose into 3-simplices. A two-dimensional analogue is depicted in Fig. 2. A particular manifold can be triangulated in many different ways. It turns out that any two different triangulations of a same underlying 2-manifold can be

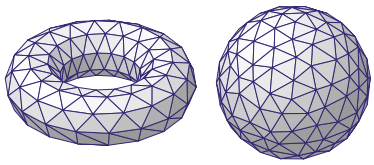


Figure 2: Triangulation of surfaces.

related to each other by a series of elementary modifications known as two-dimensional Pachner moves, which are shown in Fig. 3. Thus, if we want to construct a manifold invariant for a



Figure 3: 1-3 and 2-2 Pachner moves in two dimensions.

2-manifold represented in terms of a triangulation, we only need to find some function of the triangulation that is invariant under these two Pachner moves.

The story is quite similar in three dimensions. We discretize any closed 3-dimensional manifold by breaking it up into tetrahedra (three-dimensional simplices). Any two discretizations are topologically equivalent to each other if and only if they can be related to each other by a series of three-dimensional Pachner moves, which are shown in Fig. 4. Again, the key point here is that if we can find some function of the network structure that is invariant under the Pachner moves, we will have constructed a topological invariant of the manifold.

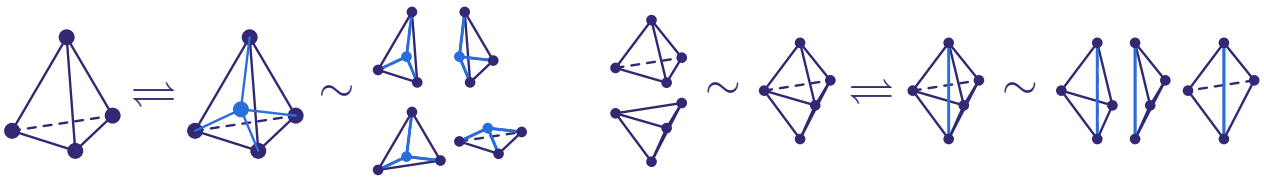


Figure 4: The 1-4 and 2-3 Pachner moves in three dimensions.

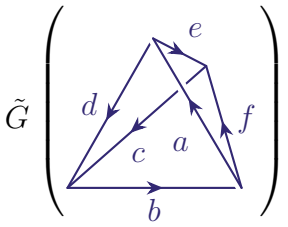
2.2.2 Turaev–Viro theories

For simplicity, we assume that $N_{ab}^c = 0$ or 1. Given a planar diagram algebra, the associated Turaev–Viro theory assigns a partition function to a 3-manifold $\mathcal{M}$ by

$$\begin{aligned} Z_{\mathrm{TV}}(\mathcal{M}) &= \mathcal{D}^{-2N_v} \sum_{\text{labelings}} W(\text{labeling}), \\ \mathcal{D}^2 &= \sum_a d_a^2, \\ W(\text{labeling}) &= \frac{\prod_{\text{tetrahedra}} \tilde{G}(\text{tetrahedron}) \prod_{\text{edges}} d_{\text{edge}}}{\prod_{\text{triangles}} \tilde{\Theta}(\text{triangle})}, \end{aligned} \tag{9}$$

where d_a is the quantum dimension of particle type a and $\tilde{G}, \tilde{\Theta}$ are as given by

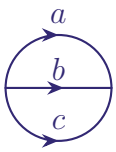
$$\tilde{\Theta}\left(\quad\right) = \Theta(a,b,c) = \sqrt{d_a d_b d_c}$$



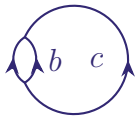
$$\tilde{G}\left(\quad\right) = G^{bad}_{ecf} = F^{bad}_{ecf}\, d_f \sqrt{\frac{d_b d_c}{d_f}} \sqrt{\frac{d_a d_e}{d_f}}$$

Here, the numbers $\Theta(a,b,c)$ and G^{bad}_{ecf} are determined by evaluating the following planar diagrams:

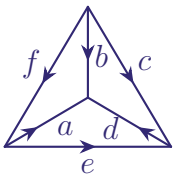
$$\Theta(a,b,c) \;=\;$$



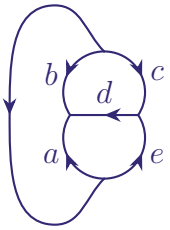
$$= a$$



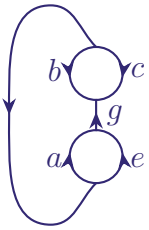
$$= d_c \sqrt{\frac{d_a d_b}{d_c}} = \sqrt{d_a d_b d_c}$$



$$= f$$

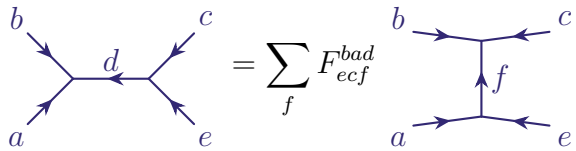


$$= \sum_g F^{bad}_{ecg}\; f$$



$$= F^{bad}_{ecf}\, d_f \sqrt{\frac{d_b d_c}{d_f}} \sqrt{\frac{d_a d_e}{d_f}} \equiv G^{bad}_{ecf}$$

Remark 2.3. *The convention for the F-matrices in the above definition of G differs from the F-matrices in Eq. (8). Here, the symbol F^{bad}_{ecf} is defined by the following diagrammatic relation,*



and related to the F-matrices in Eq. (8) by

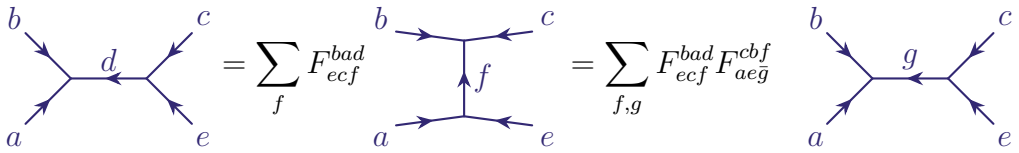
$$F^{bad}_{ecf} = [F^{\bar{a}\bar{b}\bar{c}}_{e}]_{df}.$$

Note that rotating the above diagram by 180 degrees and comparing the original diagram, we get the constraint

$$F^{bad}_{ecf} = F^{ec\bar{d}}_{ba\bar{f}}. \tag{10}$$

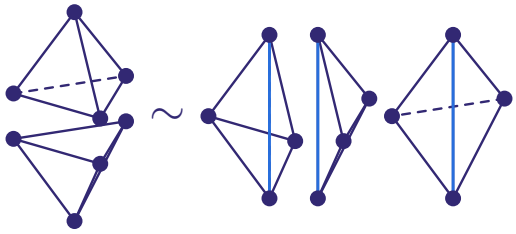
Furthermore, we can apply the F-matrix twice and get

$$\sum_f F^{bad}_{ecf} F^{cbf}_{ae\bar{g}} = \delta_{dg}. \tag{11}$$



To show that $Z_{\mathrm{TV}}(\mathcal{M})$ is a manifold invariant, one only needs to show that it is invariant under the 1-4 and 2-3 Pachner moves. For example, invariance under the 2-3 Pachner move is equivalent to the pentagon equation.

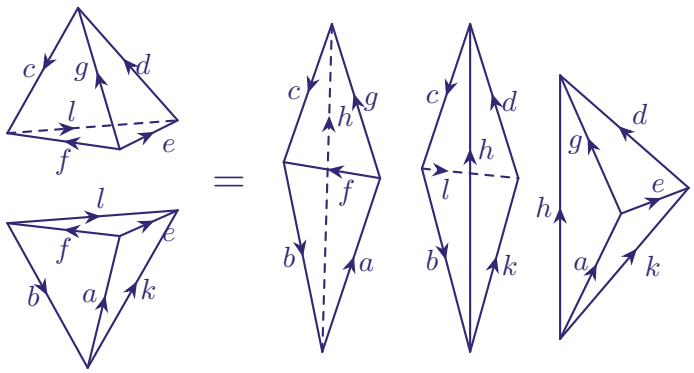
Proposition 2.4. *$Z_{\mathrm{TV}}(\mathcal{M})$ is invariant under the 2-3 Pachner move.*



Proof. On the left we have two tetrahedra (call them 1 and 2), joined along a horizontal triangle (call it α). On the right we have three tetrahedra (call them 3, 4, 5) joined along three vertical triangles (call them β, γ, δ). The three vertical triangles intersect along a vertical edge, which we label by the particle type n. To show that Z_{TV} remains invariant we need to show that

$$\frac{\tilde{G}(1)\tilde{G}(2)}{\tilde{\Theta}(\alpha)} = \sum_n \frac{\tilde{G}(3)\tilde{G}(4)\tilde{G}(5) d_n}{\tilde{\Theta}(\beta)\tilde{\Theta}(\gamma)\tilde{\Theta}(\delta)}. \tag{12}$$

Using the following labelings,



and invoking $d_x = d_{\bar{x}}$, Eq. (12) reads

$$F^{c\bar{f}g}_{edl} F^{baf}_{e\bar{l}k} \sqrt{\frac{(d_c d_d d_f d_e)(d_b d_l d_a d_e)}{d_e d_l d_f}} = \sum_h F^{baf}_{gch} F^{\bar{h}ag}_{edk} F^{cbh}_{kdl}\, d_h \sqrt{\frac{(d_b d_c d_a d_g)(d_h d_d d_a d_e)(d_c d_d d_b d_k)}{(d_h d_c d_b)(d_h d_k d_d)(d_a d_g d_h)}}.$$

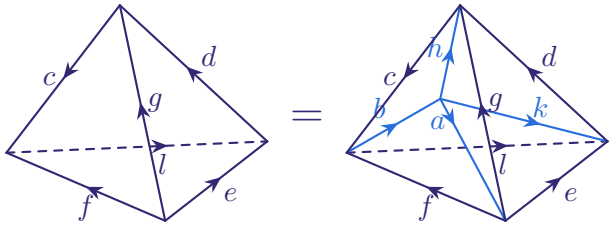
The quantum dimensions d_x completely cancel out, and the pentagon equation remains:

$$F^{c\bar{f}g}_{edl} F^{baf}_{e\bar{l}k} = \sum_h F^{baf}_{gch} F^{\bar{h}ag}_{edk} F^{cbh}_{kdl}. \tag{13}$$

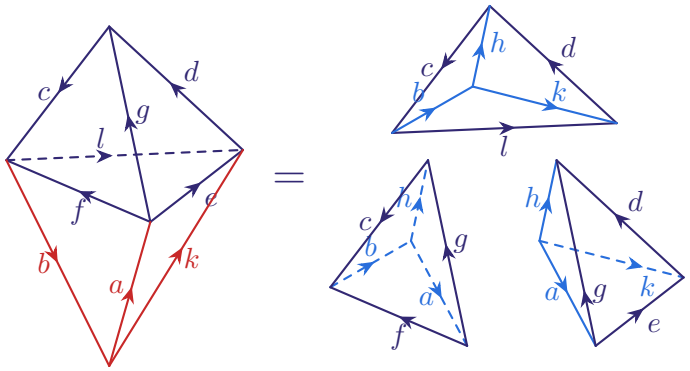
□

Proposition 2.5. *$Z_{\mathrm{TV}}(\mathcal{M})$ is invariant under the 1-4 Pachner move.*

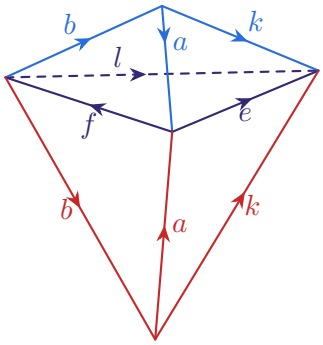
Proof. What we want to prove can be represented diagrammatically as follows.



First computing the upper three tetrahedra in the right hand side, and summing over the edge h, we get the following.



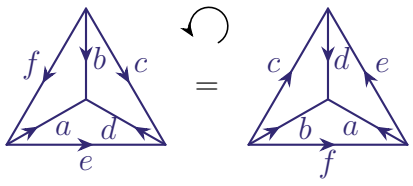
This is precisely the desired large tetrahedron times the following,



multiplied by $d_a d_b d_k$ and divided by $\mathcal{D}^2\tilde{\Theta}(\bar{a},\bar{b},\bar{f})\tilde{\Theta}(\bar{k},a,e)\tilde{\Theta}(b,k,\bar{l})\tilde{\Theta}(f,l,\bar{e})$, and summed over a,b,k. This deviation from the large tetrahedron is precisely

$$J = \mathcal{D}^{-2}\sum_{a,b,k} G^{baf}_{e\bar{l}k} G^{\bar{b}\bar{f}\bar{a}}_{e\bar{k}l} \frac{1}{d_e d_f d_l}.$$

Using the symmetry property of the symbol G induced from the $2\pi/3$ rotation of triangle,



namely $G^{bad}_{ecf} = G^{dba}_{f\bar{e}\bar{c}}$, and also using Eq. (10), we compute

$$\begin{aligned}\mathcal{D}^2 J &= \sum_{a,b,k} G^{baf}_{e\bar{l}k} G^{e\bar{k}a}_{\bar{b}f\bar{l}} \frac{1}{d_e d_f d_l} = \sum_{a,b,k} G^{baf}_{e\bar{l}k} G^{ae\bar{k}}_{\bar{l}bf} \frac{1}{d_e d_f d_l} = \sum_{a,b,k} G^{baf}_{e\bar{l}k} G^{\bar{l}bk}_{ae\bar{f}} \frac{1}{d_e d_f d_l} \\ &= \sum_{a,b,k} F^{baf}_{e\bar{l}k} F^{\bar{l}bk}_{ae\bar{f}} \frac{d_a d_b d_e d_l}{d_e d_f d_l} = \sum_{a,b} \frac{d_a d_b}{d_f} \sum_k F^{baf}_{e\bar{l}k} F^{\bar{l}bk}_{ae\bar{f}} = \sum_{a,b} \frac{d_a d_b}{d_f}. \end{aligned} \tag{14}$$

In the last equality, $\sum_k F^{baf}_{e\bar{l}k} F^{\bar{l}bk}_{ae\bar{f}} = 1$ by Eq. (11) when the vertex formed by a, b, f is admissible, while the sum vanishes otherwise. Hence the sum over a, b in Eq. (14) runs only over admissible labels; writing this constraint explicitly with $N^f_{ab} \in \{0, 1\}$,

$$\mathcal{D}^2 J = \frac{1}{d_f} \sum_{a,b} N^f_{ab} d_a d_b = \mathcal{D}^2, \tag{15}$$

where we invoked Proposition 2.1. As a result, the deviation J is a unity, and we get the desired result. □

For manifold with boundary, $\partial\mathcal{M} \neq \emptyset$, we denote by $a_1, \ldots, a_N$ a labeling of the edges on the boundary, and set the corresponding wave function

$$Z_{\mathrm{TV}}(\mathcal{M}; a_1, \ldots, a_N) = \mathcal{D}^{-2N_v - n_v} W'(a_1, \ldots, a_N) \sum_{\text{bulk labelings}} W(\text{bulk label}) \tag{16}$$

where N_v is the number of vertices in the bulk and n_v is the number of vertices on the boundary. The bulk weight W is exactly the same as in Eq. (9). An additional weight is included, which is a function of the boundary labels

$$W'(a_1, \ldots, a_N) = \left(\frac{\prod_{\text{edges}} d_{\text{edge}}}{\prod_{\text{triangles}} \tilde{\Theta}(\text{triangle})} \right)^{1/2}. \tag{17}$$

With Eq. (9) and Eq. (17), one has the gluing equality for a pair of manifolds with boundary $\mathcal{M}, \mathcal{M}'$ with $\partial\mathcal{M} = \partial\mathcal{M}' = \Sigma$.

$$\begin{aligned} &Z_{\mathrm{TV}}(\mathcal{M} \cup_\Sigma \mathcal{M}') = \langle Z_{\mathrm{TV}}(\mathcal{M}') | Z_{\mathrm{TV}}(\mathcal{M}) \rangle \\ &= \sum_{a_1, \ldots, a_N} [Z_{\mathrm{TV}}(\mathcal{M}'; \bar{a}_1, \ldots, \bar{a}_N)]^* Z_{\mathrm{TV}}(\mathcal{M}; a_1, \ldots, a_N) \qquad (18a) \\ &= \sum_{a_1, \ldots, a_N} Z_{\mathrm{TV}}(\mathcal{M}'; a_1, \ldots, a_N) Z_{\mathrm{TV}}(\mathcal{M}; a_1, \ldots, a_N) \qquad (18b) \end{aligned}$$

2.3 Dijkgraaf–Witten

2.3.1 Group cohomology

Let G be a group, A an abelian group, and $F_n = \{f : G^n \to A\}$ be the set of A-valued n-ary functions on G. F_n is naturally an abelian group. We note that an element of $G^n = \underbrace{G \times \cdots \times G}_{n\text{-fold}}$ can be represented by an n-simplex, so that the elements of F_n are functions on the set of n-simplices. To this end, let $(g_1, g_2, \ldots, g_n)$ be given. We start from a concatenated set of n arrows, labeled by $g_1, g_2, \ldots, g_n$, respectively (see Fig. 5).

Next, we connect from the initial point of g_i and the final point of g_{i+1} and label this new arrow by $g_{i+1}g_i$. Next, we do the same for new pairs of arrows by adjoining the initial and final point of existing arrows. This ends in a finite step, and the resulting diagram is a n-dimensional simplex whose edges are labeled by elements of $g_1, \ldots, g_n$ and their compositions.

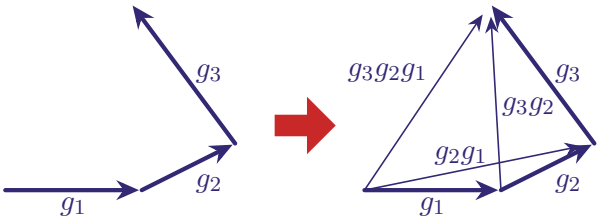


Figure 5: Concatenated arrows that determine a 3-simplex

We can define a coboundary operator $\delta : F_n \to F_{n+1}$ as follows. Let $\alpha \in F_n$ be given. The boundary of a simplex $\Delta_{n+1} = (g_1, g_2, \ldots, g_{n+1})$ belonging to G^{n+1} is composed of $n+2$ distinct n-simplices, on which we can evaluate α. We define $\delta\alpha$ by

$$(\delta\alpha)(\Delta_{n+1}) = \sum_{\Delta_n \subset \partial\Delta_{n+1}} \mathrm{sgn}(\Delta_n)\alpha(\Delta_n) \tag{19}$$

where $\mathrm{sgn}(\Delta_n) = +1$ if the orientation of Δ_n is such that if the ordered tuple $(\Delta_{n+1}\backslash\Delta_n, \Delta_n) \in G^{n+1}$ is an even permutation of Δ_{n+1}, and $\mathrm{sgn}(\Delta_n) = -1$ otherwise. Here, $\Delta_{n+1}\backslash\Delta_n$ denotes the only point in Δ_{n+1} that does not belong to Δ_n. One can check that $\delta\delta : F_n \to F_{n+2}$ is zero for all n.

The d-th group cohomology $H^d(G, A)$ is defined to be the cohomology group $\ker\delta/\operatorname{im}\delta$ of the chain complex

$$A \cong F_0 \xrightarrow{\delta} F_1 \xrightarrow{\delta} F_2 \xrightarrow{\delta} F_3 \xrightarrow{\delta} \cdots$$

taken at F_d.

Example 2.6. *The group cocycle condition for the first group cohomology $H^1(G, \mathrm{U}(1))$ is*

$$\omega_1(g)\omega_1(h) = \omega_1(gh), \tag{20}$$

which means that ω_1 is a group homomorphism. Since $\operatorname{im}\delta = 0$ at F_1, $H^1(G, A)$ is just homomorphisms from G to A. Note that the group operation is multiplicative since $A = \mathrm{U}(1)$.

Example 2.7. *The group cocycle condition for the second group cohomology $H^2(G, \mathrm{U}(1))$ is*

$$\omega_2(g, h)\omega_2(gh, k) = \omega_2(h, k)\omega_2(g, hk) \tag{21}$$

which is equivalent to Fig. 6. Eq. (21) states that a partition function constructed from ω_2 is invariant under the 2-2 Pachner move.

Remark 2.8. *Examples 2.6, 2.7 are actually a special case of group cohomology, where G trivially acts on A. More generally, group cohomology is defined when A is a G-module. This deforms the cohomology groups, and $H^1(G, A)$ would not simply be group homomorphisms anymore.*

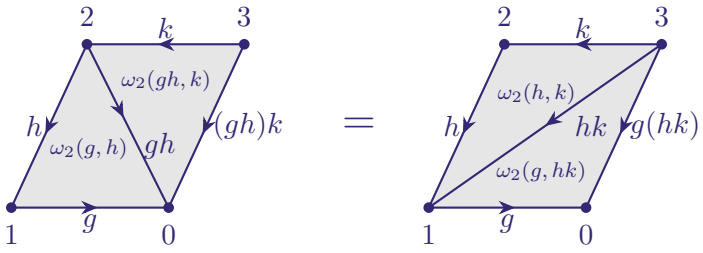


Figure 6: Diagram representing the 2-cocycle condition of group cohomology.

2.3.2 Dijkgraaf–Witten theories

Now, let G be a finite group. A d-cocycle $\omega_d \in Z^d(G, \mathrm{U}(1))$ defines a Dijkgraaf-Witten TQFT in d dimensions by

$$Z_{\mathrm{DW}}(\mathcal{M}) = |G|^{-N_v} \sum_{\text{labelings}} \prod_{\text{simplices}} \omega_d(g_{j_2 j_1}, \ldots, g_{j_{d+1} j_d})^{s(j_1,\ldots,j_{d+1})} \tag{22}$$

where $j_1, \ldots, j_{d+1}$ are vertices, $g_{jj'}$ the label of the edge connecting j' to j, and $s(j_1, \ldots, j_{d+1}) = \pm 1$ the relative orientation of the simplex $[j_1, \ldots, j_{d+1}]$ with respect to the underlying d-manifold. The cocycle condition guarantees that Eq. (22) is invariant under the Pachner moves. Also, if $\omega_d = \delta\eta_{d-1}$ is a coboundary, the term $\omega_d(g_{j_2 j_1}, \ldots, g_{j_{d+1} j_d})$ is solely defined by the boundary $(d-1)$-simplices, and each $(d-1)$-simplex appears twice in Eq. (22) since $\mathcal{M}$ is closed, because in a triangulation of a closed d-manifold each $(d-1)$-simplex lies on the boundary of exactly two d-simplices. Also, the two instances of a$(d-1)$-simplex appear with opposite orientation. This proves that a Dijkgraaf–Witten theory is in fact determined by a choice of group cohomology element $[\omega_d] \in H^d(G, \mathrm{U}(1))$.

Remark 2.9. *One particularly interesting special case of Dijkgraaf–Witten theory is the case of the trivial d-cocycle where ω_d is always unity. In this case, Eq. (22) just counts the number of flat field configurations and then divides by $|G|^{N_v}$. This partition function is exactly that of lattice gauge theory, and the resulting topological quantum field theory is known as the quantum double of the group G. The more general case, with a nontrivial cocycle, is correspondingly sometimes known as "twisted" gauge theory, where the cocycle is thought of as some sort of twist to the otherwise simple theory. In this regard, Dijkgraaf–Witten theories are sometimes called twisted quantum doubles.*

References

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[2] Steven H. Simon. *Topological Quantum.* Oxford University Press, Sept. 2023. ISBN: 9780191994425. DOI: 10.1093/oso/9780198886723.001.0001.